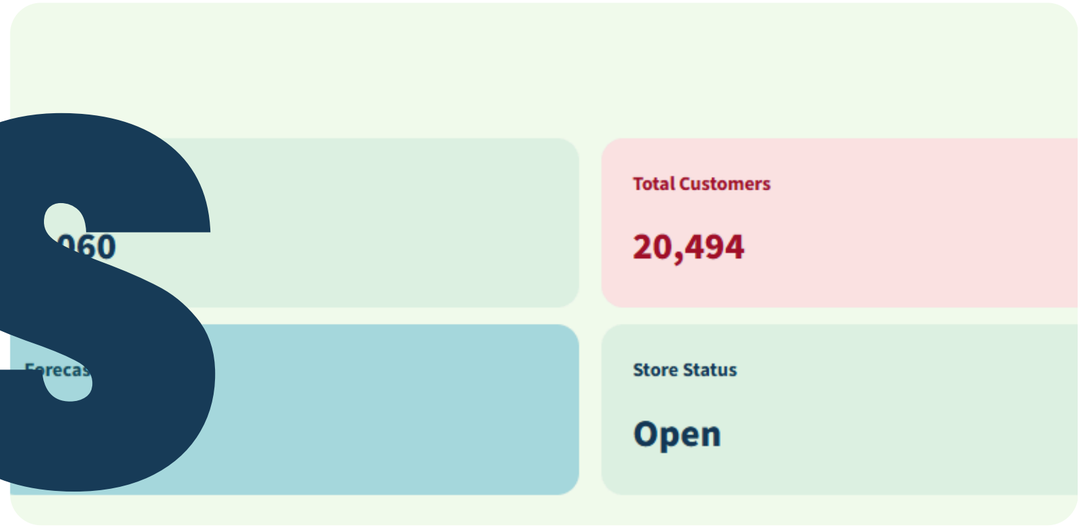
1ST NOVEMBER 2026

ROSSMAN SALES



BY SIDDHEYA LOHAR

OBJECTIVES

Project objectives:

1. Analyze historical sales and customer behaviour.
2. Identify factors influencing retail performance.
3. Develop machine learning models to forecast sales and customers.
4. Evaluate models using appropriate performance metrics.
5. Build an interactive forecasting application.

Key challenges:

- Fluctuating daily sales.
- Seasonal demand variations.
- Promotion and holiday effects.
- Differences in customer traffic across stores.
- Competition between retail locations.

DATASET OVERVIEW

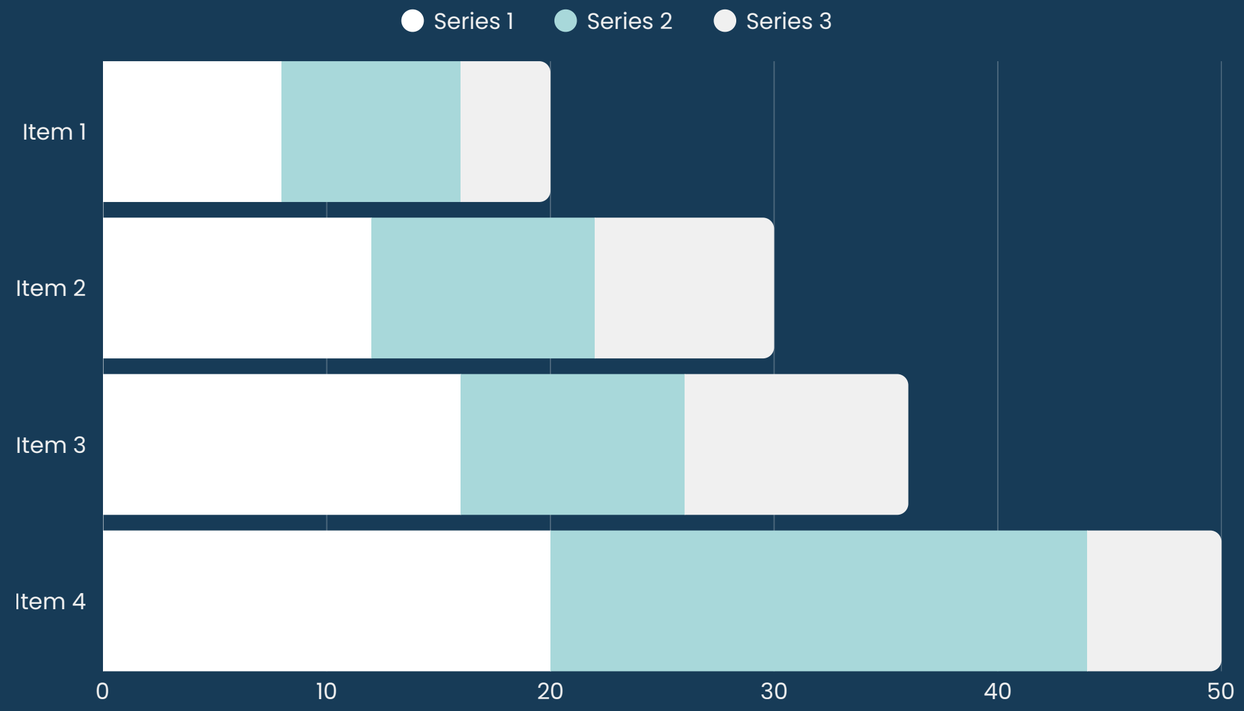
DATA STATS

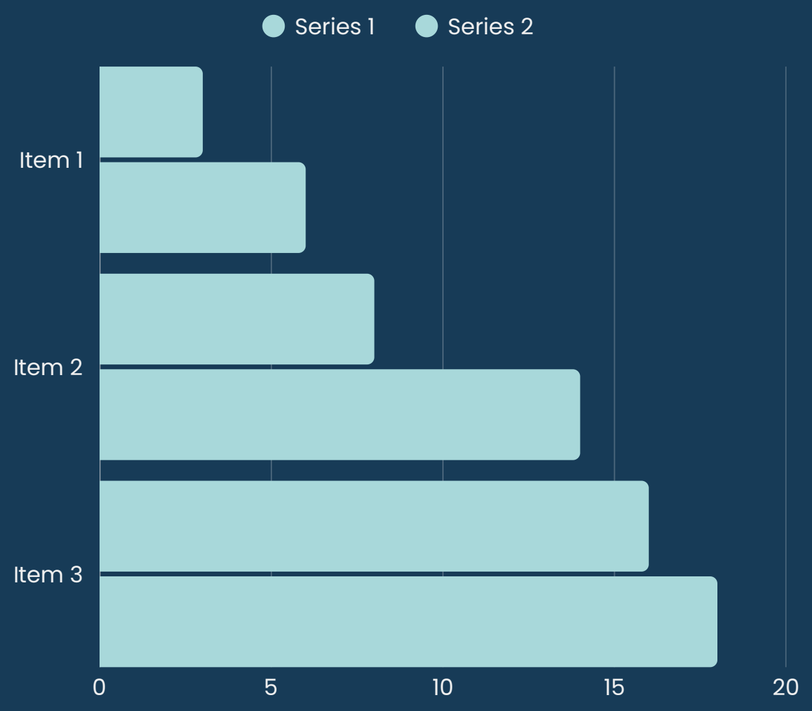
Training dataset
1,017,209
Testing dataset
41,088
Store metadata
1,115 stores
Training period
Jan 2013 – Jul 2015
Testing period
Aug 2015 – Sep 2015

	Id	Store	DayOfWeek	Date	Open	Promo	StateHoliday	SchoolHoliday
479	480	622	4	2015-09-17	NaN	1	0	0
1335	1336	622	3	2015-09-16	NaN	1	0	0
2191	2192	622	2	2015-09-15	NaN	1	0	0
3047	3048	622	1	2015-09-14	NaN	1	0	0
4759	4760	622	6	2015-09-12	NaN	0	0	0
5615	5616	622	5	2015-09-11	NaN	0	0	0
6471	6472	622	4	2015-09-10	NaN	0	0	0
7327	7328	622	3	2015-09-09	NaN	0	0	0
8183	8184	622	2	2015-09-08	NaN	0	0	0
9039	9040	622	1	2015-09-07	NaN	0	0	0
10751	10752	622	6	2015-09-05	NaN	0	0	0

FEATURES

Sales
Customers
Promo
StateHoliday
Date

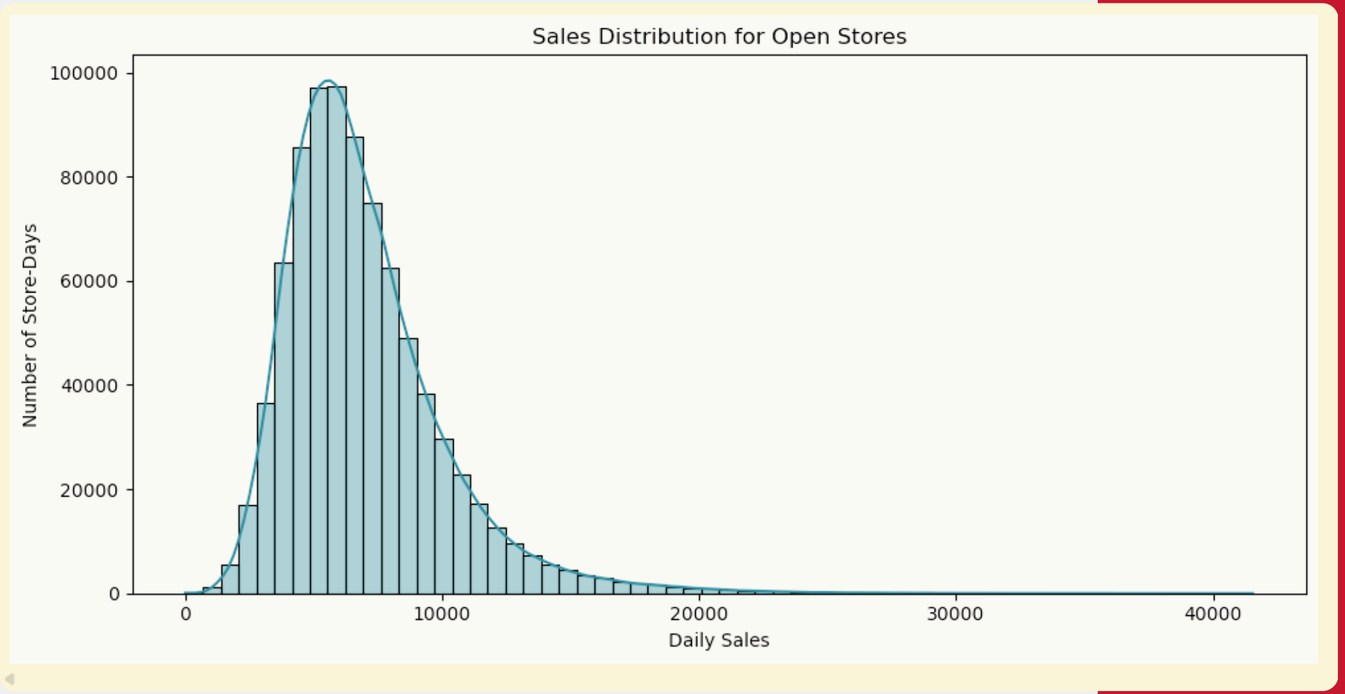


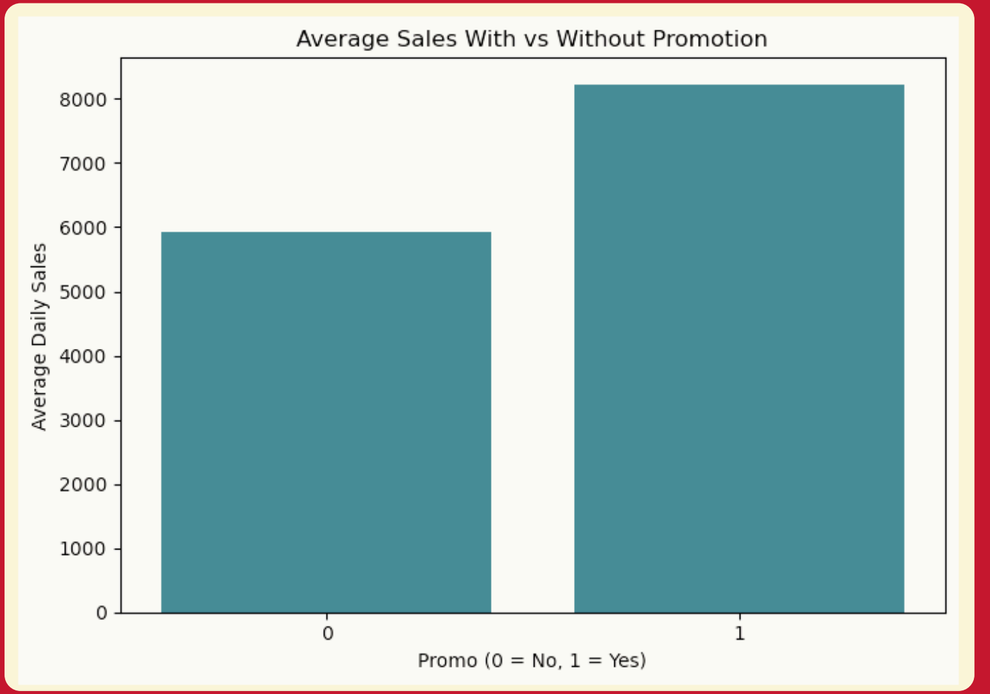


EDA

Discovering Sales Patterns

GRAPHS


Sales Distribution for Open Stores
Number of Store-Days
Daily Sales
0
20000
40000
60000
80000
100000
0
10000
20000
30000
40000


Average Sales With vs Without Promotion
Average Daily Sales
Promo (0 = No, 1 = Yes)
0
1000
2000
3000
4000
5000
6000
7000
8000
0
1

FEATURE ENGINEERING

- Combined sales and store metadata.
- Handled missing values.
- Processed categorical variables.
- Prepared training and validation datasets.
- Maintained a chronological validation split.

Feature	Purpose
Year	Annual trends
Month	Seasonal patterns
Day	Daily variations
DayOfWeek	Weekly patterns
WeekOfYear	Weekly seasonality
IsWeekend	Weekend effects
IsMonthStart	Beginning-of-month effects
IsMonthEnd	End-of-month effects
Competition features	Competitive environment
Promotion features	Promotional activity

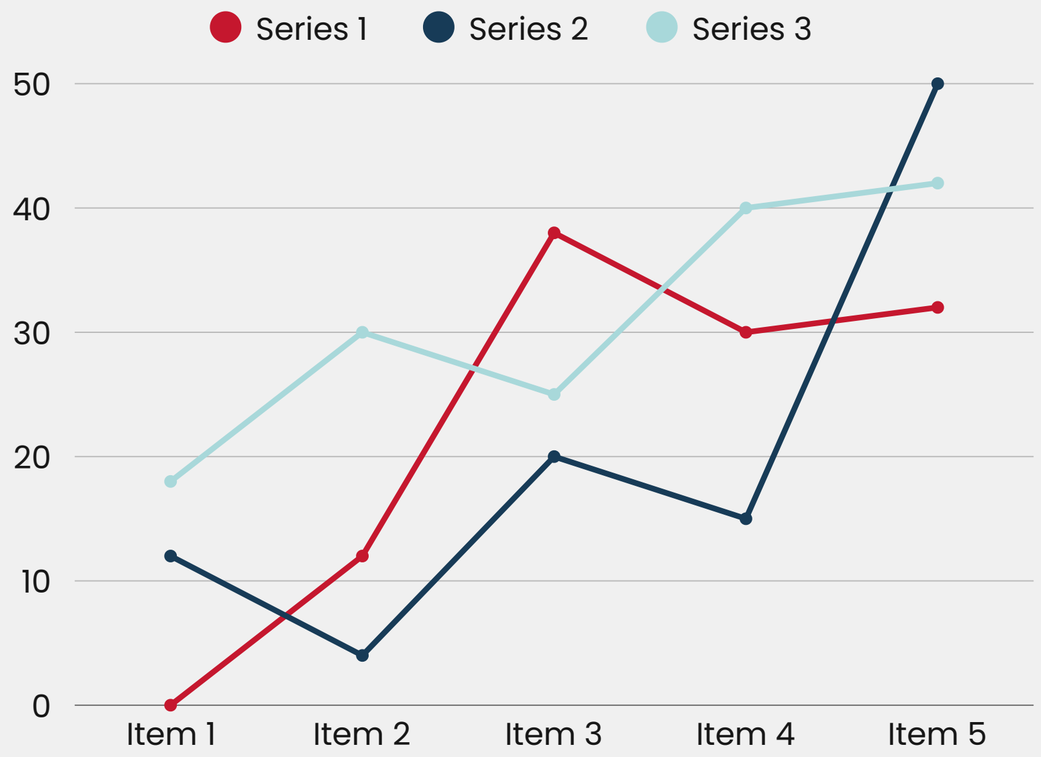
MACHINE LEARNING

MODELS EXPLOERED

- Store-average baseline model.
- Random Forest Regressor.
- Extra Trees Regressor.
- Customer forecasting model using Extra Trees.
- Global LSTM for time-series forecasting.

Model	MAE	RMSE
Store Average Baseline	1,486.65	1,979.55
Random Forest	710.71	1,056.97
Extra Trees	**705.14**	**1,021.69**

KEY RESULT



- **MAE: 55.56**
- **RMSE: 79.40**

LSTM:

- Developed a global time-series forecasting model.
- Saved the trained model and preprocessing artifacts.

Evaluation metrics:

- MAE: Average absolute prediction error.
- RMSE: Penalizes larger prediction errors more heavily.



MLflow – Experiment Tracking

- Recorded model training experiments.
- Tracked model performance and parameters.
- Used SQLite-backed experiment storage.
- Maintained experiment history.



DVC – Data Version Control

- Initialized DVC for dataset tracking.
- Tracked training, testing and store datasets.
- Created Git-tracked dataset snapshots.
- Demonstrated dataset version changes.



Streamlit – Interactive Application

Developed an interactive forecasting dashboard with:

- Store selection.
- Custom date-range selection.
- Promotion and holiday controls.
- Sales and customer predictions.
- Performance visualizations.
- CSV forecast downloads.

EXPERIMENTAL TRACKING

DATA VERSIONING

DEPLOYMENT

RESULTS & CONCLUSION

Project achievements:

- Processed over one million training records.
- Identified important sales and customer patterns.
- Improved validation MAE by approximately 52.6% over the baseline.
- Developed separate sales and customer forecasting models.
- Explored global LSTM forecasting.
- Implemented MLflow experiment tracking.
- Established DVC dataset versioning.
- Built an interactive Streamlit application.

Business applications:

- Better inventory planning.
- Improved staffing decisions.
- Promotion planning.
- Understanding customer traffic.
- Supporting store-level business decisions.

Future improvements:

- More advanced time-series models.
- Improved holiday and promotion forecasting.
- Automated model retraining.
- Integration with live retail data.

THANK YOU